Policy brief

HAI PHONG EXPLORES AN HPV TESTING CO-PAYMENT MODEL FOR CERVICAL CANCER SCREENING



July 2026

> *With support from FIND, the Hai Phong Department of Health has developed a co-payment model for HPV DNA testing. The proposed model is informed by local evidence generated from Hai Phong, including primary and secondary data, as well as extensive stakeholder consultations. It is intended to support evidence-based policy recommendations for consideration by the city authorities at an appropriate stage of the policy-making process.*

EXECUTIVE SUMMARY

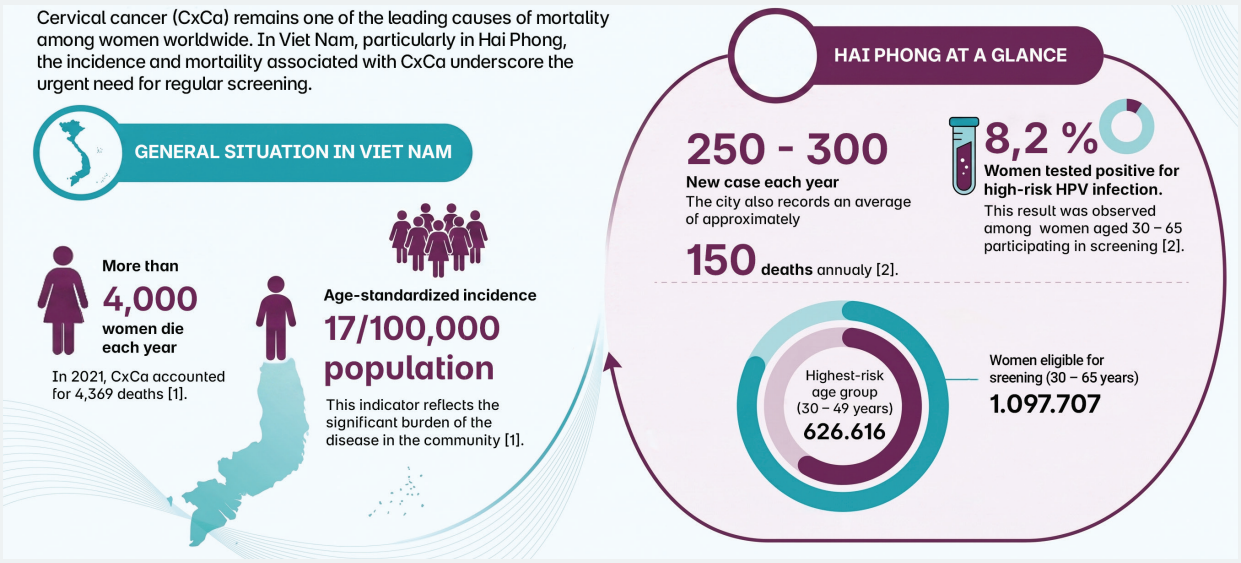
Cervical cancer (CxCa) remains one of the most common cancers affecting women. Although the World Health Organization (WHO) recommends HPV DNA testing as the preferred, first-choice method for CxCa screening, access to screening services in Viet Nam—including Hai Phong city remains limited. Key barriers include policy gaps, financial constraints, and the absence of a reimbursement scheme for preventive screening services under the Social Health Insurance (SHI).

Between 2023 and 2024, with financial and technical support from FIND and under the leadership of the Hai Phong Department of Health, the Hai Phong Center for Disease Control (CDC) and primary healthcare facilities piloted a model of decentralized HPV DNA sample collection at Commune Health Stations (CHSs), combined with centralized laboratory testing at the CDC. The pilot demonstrated that the model is operationally feasible, well accepted by both communities and the health system, and provides a strong foundation for scale-up once an appropriate financing and policy framework is established.

Building on this success and at the request of the Hai Phong Department of Health, FIND has continued to support the development of an HPV DNA Testing Co-payment Model in Viet Nam. The model presents six financing scenarios that combine contributions from city's budget source and service users. It also outlines payment arrangements, financial flow management mechanisms, implementation processes, financial and programme monitoring frameworks, and a phased implementation roadmap for 2026–2031. In addition, the model assesses the expected health and socioeconomic benefits of scaling HPV DNA screening.

Hai Phong has many advantages to become one of the first cities in Viet Nam to implement the co-payment model generating practical experience and scientific evidence to inform the development of sustainable financing, management and operational mechanisms. It is also expected to draw lessons to support the future inclusion of HPV DNA testing within the Social Health Insurance and facilitate nationwide scale-up.

BURDEN OF CERVICAL CANCER



FINANCING GAP

- HPV DNA testing is a highly effective method for CxCa screening and early detection and is recommended by both the WHO and the Ministry of Health of Viet Nam. However, under the current financing mechanism, the SHI scheme does not reimburse HPV DNA testing for routine screening.
- With more than one million women eligible for CxCa screening, financing a population-based screening programme represents a substantial fiscal challenge for the local budget. Therefore, a co-payment mechanism that service users share costs offers a practical financing approach.
- The proposed co-payment model is intended as an interim financing solution to expand access to HPV DNA screening while generating real-world evidence to inform the development of a SHI reimbursement mechanism for the service.

750.000 - 1.250.000
VND / test

The testing cost remains high and is primarily covered for treatment purposes. This prompts individuals to seek out and use the service out-of-pocket, though under heavily restricted conditions.

RECOMMENDED CxCa SCREENING APPROACH

The WHO recommends HPV DNA testing as the first-hand screening method for women aged 30 years and older, with screening intervals of 5–10 years. When women have access to regular screening services, the vast majority of CxCa cases can be prevented through early detection and timely management of precancerous lesions [3].

POLICY CONTEXT

The WHO has set the global goal of eliminating CxCa as a public health problem through the 90–70–90 strategy by 2030: 90% of girls fully vaccinated against HPV by age 15, 70% of women screened with a high-performance test by age 35 and again by age 45, and 90% of women identified with cervical disease receiving appropriate treatment.

In Viet Nam, the Government and the Ministry of Health have issued a range of policies to strengthen CxCa prevention, early detection, and control, including Resolution No. 72-NQ/TW, Resolution No. 262/2025/QH15, and Decision No. 358/QD-BYT.

At the local level, the Hai Phong Party Committee and People's Committee have identified preventive healthcare and population health as strategic priorities for sustainable socioeconomic development, as reflected in Resolution No. 05-NQ/TU.

GUIDING PRINCIPLES FOR THE CO-PAYMENT MODEL

Designed for population-based screening: The proposed model is intended to support large-scale, organized CxCa screening programmes.

Promoting financial equity: Costs are shared between the local government budget and service users, while ensuring priority support for poor households, social assistance beneficiaries, and other vulnerable groups, thereby promoting equitable access to cervical cancer screening services.

Ensuring financial sustainability: By combining stable public financing with individual contributions, the model aims to establish a sustainable funding mechanism and reduce reliance on short-term donor support.

Laying the foundation for SHI integration: The proposed co-payment levels are aligned with the principles of the SHI system, providing scientific and practical evidence to support the gradual inclusion of HPV DNA testing within the SHI benefit package.

ESTIMATED COST OF HPV DNA TESTING FOR POPULATION-BASED SCREENING

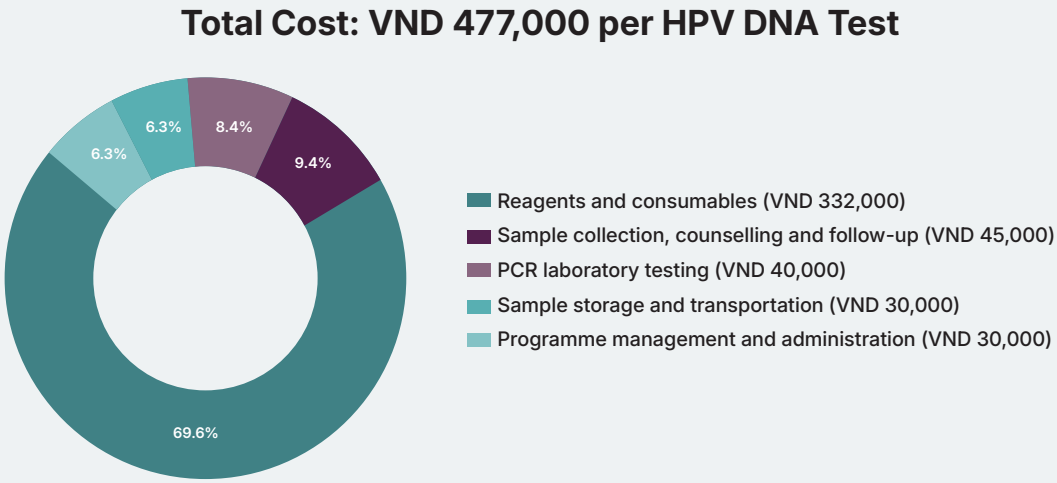


Figure 1. Cost Components of an HPV DNA Test

The proposed unit cost of VND 477,000 per HPV DNA test was estimated based on the pricing framework and cost components specified in regulations issued by the Ministry of Health for public healthcare facilities, with appropriate adjustments to reflect the conditions of population-based screening.

The estimated cost comprises reagents and consumables (including sample collection kits), which are covered under a negotiated procurement package with suppliers. The package also includes equipment depreciation through a reagent rental arrangement, whereby laboratory equipment is installed and maintained by the suppliers. This approach minimizes upfront capital investment and reduces equipment operation and maintenance costs for testing facilities.

PROPOSED SCREENING SCENARIOS

Based on available financial resources and target screening coverage, Hai Phong may consider one of the following two population-based screening scales:

SCALE A: Screening for all women aged 30–65 year

This screening scale targets approximately 1.1 million women aged 30–65 years. With a proposed screening coverage of 82%, around 900,000 women would be screened over a five-year cycle, equivalent to approximately 180,000 HPV DNA tests annually.

Scale A is appropriate when sufficient financial resources are available, and comprehensive population coverage is prioritized to accelerate progress towards CxCa elimination. This option is expected to deliver the greatest long-term public health impact.

SCALE B: Prioritized Screening for women aged 30–49 Years

This screening scale targets approximately 627,000 women aged 30–49 years, the population at highest risk of developing CxCa. Assuming the same target coverage of 82%, approximately 514,000 women would be screened over a five-year cycle, equivalent to around 103,000 HPV DNA tests annually.

Scale B requires a lower level of financial investment, making it suitable for the initial phase of implementation or in resource-constrained settings. It is expected to provide the highest cost-effectiveness while establishing a foundation for progressively expanding screening coverage in subsequent phases.

PROPOSED CO-PAYMENT SCENARIOS

Three co-payment scenarios (Scenarios 1–3) are proposed for Scale A, and three additional scenarios (Scenarios 4–6) are proposed for Scale B. Within each screening scale, 3 scenarios differ according to the level of financial contribution by service users: Scenarios 1 and 4: Co-payment levels aligned with the current SHI cost-sharing mechanism; Scenarios 2 and 5: User co-payment increased by 15 percentage points compared with the current SHI cost-sharing mechanism; Scenarios 3 and 6: User co-payment increased by 30 percentage points compared with the current SHI cost-sharing mechanism. Table 1 presents the proposed co-payment arrangements between service users and the local government per HPV DNA test, while Figure 2 illustrates the corresponding annual financing distribution between the local budget and user contributions.

Table 1. Proposed co-payment arrangements between service users and the local government per HPV DNA test

SHI Benefit Category	Scenarios 1 & 4 *Current SHI cost-sharing mechanism (VND / test)*		Scenarios 2 & 5 *User co-payment increased by 15 percentage (VND / test)*		Scenarios 3 & 6 *User co-payment increased by 30 percentage (VND / test)*	
	Service User	Local Government	Service User	Local Government	Service User	Local Government
100% SHI Coverage	0	477,000	0	477,000	0	477,000
95% SHI Coverage	23,850	453,150	95,400	381,600	166,950	310,050
80% SHI Coverage	95,400	381,600	166,950	310,050	238,500	238,500
No SHI Coverage	238,500	238,500	238,500	238,500	238,500	238,500

Across all proposed scenarios, the maximum user co-payment is capped at 50% of the test cost, equivalent to VND 238,500 per HPV DNA test. This amount remains well below the VND 400,000 per test that at least 75% of women in Hai Phong reported they would be willing to pay, according to the FIND willingness-to-pay survey [2].

Under all scenarios, individuals eligible for 100% SHI coverage, including social protection beneficiaries and other fully covered groups, are proposed to receive fully subsidized HPV DNA testing at no cost. Conversely, individuals without SHI coverage are proposed to contribute 50% of the test cost.

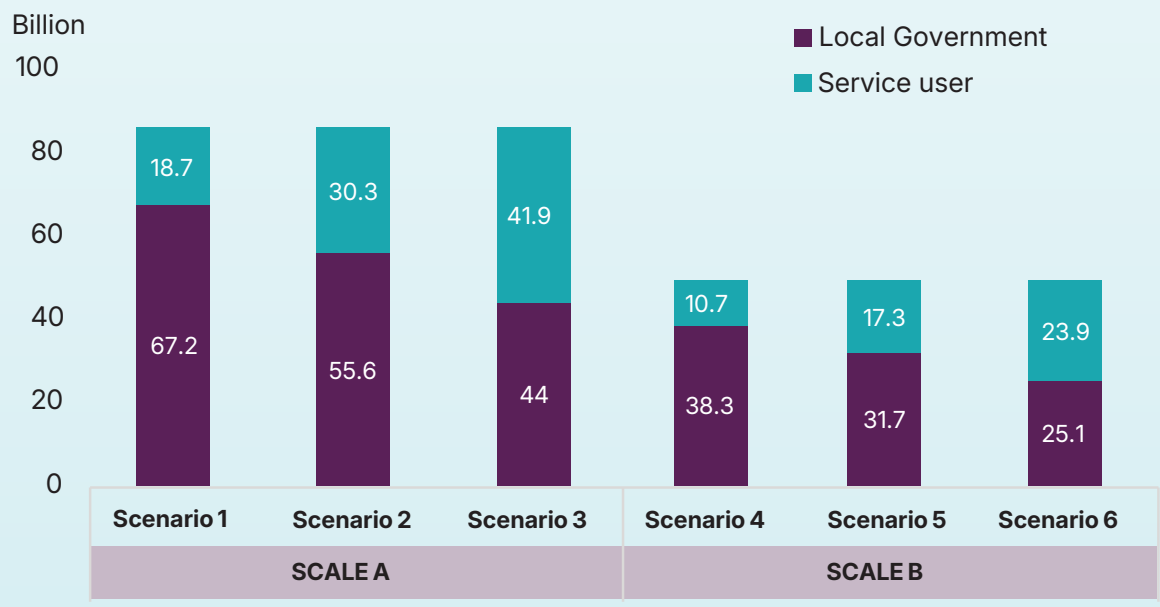


Figure 2. Estimated annual financing distribution between the local government and service users across the six co-payment scenarios (in VND)

Under Scales A and B, the estimated total annual programme cost is VND 85.9 billion and VND 49.0 billion, respectively. The annual financing requirement from the local government decreases as the user co-payment level increases. Under Scale A, the annual local government contribution is estimated to decline from VND 67.2 billion under Scenario 1 to VND 44.0 billion under Scenario 3. Under Scale B, the required local government financing is substantially lower, ranging from VND 38.3 billion under Scenario 4 to VND 25.1 billion under Scenario 6, depending on the level of user co-payment.

HEALTH IMPACT AND SOCIOECONOMIC BENEFITS

The expected impact of the proposed programme was assessed using two widely accepted health economic indicators: disability-adjusted life years (DALYs) averted, which measure the number of healthy years of life preserved, and the benefit–cost ratio (BCR), which assesses the economic return on investment. These indicators are widely used in health economic evaluations to inform evidence-based policymaking and resource allocation.

Table 2. Estimated health burden (DALYs) averted through population-based CxCa screening with HPV DNA testing in Hai Phong

INDICATOR	SCALE A	SCALE B
CxCa cases averted	1,327	767
CxCa deaths averted	717	414
Life expectancy of Vietnamese women (years) [4]	77.3	77.3
Mean age at CxCa death (years)	57	53
Years of life lost per CxCa death (L)	20.3	24.3
Disability weight (DW) [5]	0.3	0.3
Average disease duration (years)	3	3
Total years of life lost (YLL)	14,550	10,059
Total years lived with disability (YLD)	1,195	690
Total DALYs averted	15,744	10,748

Note: *YLL = Deaths averted × Years of life lost per death (L); YLD = CxCa cases averted × Disability weight (DW) × Average disease duration; DALYs averted = YLL + YLD.*

DALYs averted represent the number of healthy life years preserved through the screening programme by preventing premature deaths and reducing years lived with disability due to CxCa. As shown in Table 2, population-based HPV DNA screening has the potential to substantially reduce the burden of CxCa in Hai Phong.

Under Scale A, the programme is estimated to avert **15,744 DALYs,** equivalent to preserving **15,744 healthy life years**

Under Scale B, an estimated **10,748 DALYs** would be averted, corresponding to **10,748 healthy life years preserved**

The programme's socioeconomic benefits were estimated using the Human Capital Approach [6–7], whereby each DALY averted was assigned an economic value equivalent to Hai Phong's gross regional domestic product (GRDP) per capita, estimated at VND 194.5 million per DALY averted. This approach quantifies the economic value of preventing premature mortality, reducing the burden of disease, and maintaining workforce productivity. Table 3 presents the estimated socioeconomic benefits of implementing a population-based CxCa screening programme in Hai Phong.

Table 3. Estimated socioeconomic benefits of population-based CxCa screening in Hai Phong

INDICATOR	SCALE A	SCALE B
Cost of screening and treatment for detected precancerous lesions and CxCa (VND billion)	513.2	291.1
CxCa treatment costs averted (VND billion)	418.0	241.5
Economic value of healthy life years preserved (DALYs averted) (VND billion)	3,062.0	2,091.0
Total socioeconomic benefits (VND billion)	3,480.0	2,332.0
Net benefits (Benefits – Costs) (VND billion)	2,967.0	2,041.0
Benefit–Cost Ratio (BCR)	**6.78**	**8.01**

The economic value of DALYs averted was estimated by multiplying the number of DALYs averted by Hai Phong's estimated GRDP per capita in 2025.

Benefit–Cost Ratio (BCR) = Total socioeconomic benefits ÷ Total programme costs. Total socioeconomic benefits comprise the economic value of CxCa treatment costs averted and the economic value of DALYs averted.

The analysis indicates that population-based CxCa screening using HPV DNA testing would not only substantially reduce the burden of disease and CxCa-related mortality but also generate considerable socioeconomic returns. Both screening scales demonstrate benefits that substantially exceed programme costs. The estimated Benefit–Cost Ratio (BCR) is 6.78 for Scale A and 8.01 for Scale B. In practical terms, every VND 1 invested in the programme is expected to generate approximately VND 6.78 and VND 8.01 in socioeconomic benefits under Scales A and B, respectively.

OPERATIONAL WORKFLOW

The following operational workflow is proposed for implementing the HPV DNA screening programme within the health system:

SET 1 Community engagement and demand generation

Implement communication and community engagement activities in accordance with the communication strategy and implementation plan developed by the Department of Health and the Programme Management Unit (PMU). Activities should be conducted at multiple levels to mobilize support from the city leadership, employers, community organizations, and the public. Communication efforts should also encourage eligible women to participate in screening and help achieve the target programme coverage.

SET 2 Participant identification and eligibility assessment

CHSs identify eligible women through community clusters, workplaces, remote areas, and other priority population groups. Eligible participants are then classified according to their SHI benefit category to determine the applicable co-payment level and facilitate implementation planning.

SET 3 Co-payment collection and financial management

CHSs collect the user co-payment, where applicable, at the time of sample collection. All payments are documented, consolidated, and transferred in accordance with the programme's financial management procedures.

SET 4 Sample collection and specimen management

CHSs organize sample collection at health facilities or through outreach services. Healthcare workers provide counselling and instructions for self-collection or assist women with provider-collected sampling when needed. Each participant is assigned a unique identification code that serves as both the participant ID and specimen ID. The specimen is registered in the digital information system before being transported to the testing laboratory.

SET 5 Specimen transportation and handover

Specimens collected at CHSs are transported either directly to the provincial CDC laboratory or, where appropriate, through regional health centres for consolidated transport from remote locations, following predefined referral and logistics arrangements. Specimen transfer and receipt are documented electronically using the programme information system.

SET 6 Laboratory testing, result reporting, and payment

Testing is performed at the designated laboratory, and results are transmitted electronically to CHSs through the programme information system. Service providers reconcile service data and process payments based on electronically verified records.

SET 7 Management of HPV-positive women

CHSs provide standardized post-test counselling to ensure that women with positive HPV results receive accurate information and appropriate psychological support. Women requiring further assessment are referred and followed up in accordance with the Ministry of Health's clinical guidelines. A digital referral and case-tracking system is recommended to ensure continuity of care across all levels of the health system.

SET 8 Post-screening follow-up

CHSs maintain follow-up records through the programme information system, provide reminders for repeat screening or clinical follow-up, and ensure continuous case management while maintaining strict confidentiality of personal information.

SET 9 Monitoring, reporting, and quality improvement

Participating facilities compile programme data and submit routine reports. The Programme Management Unit conducts both remote and onsite supervision, provides technical support through a programme hotline, organizes regular review meetings, monitors programme quality, analyses programme and financial data, and uses monitoring results to inform decision-making and continuous quality improvement.

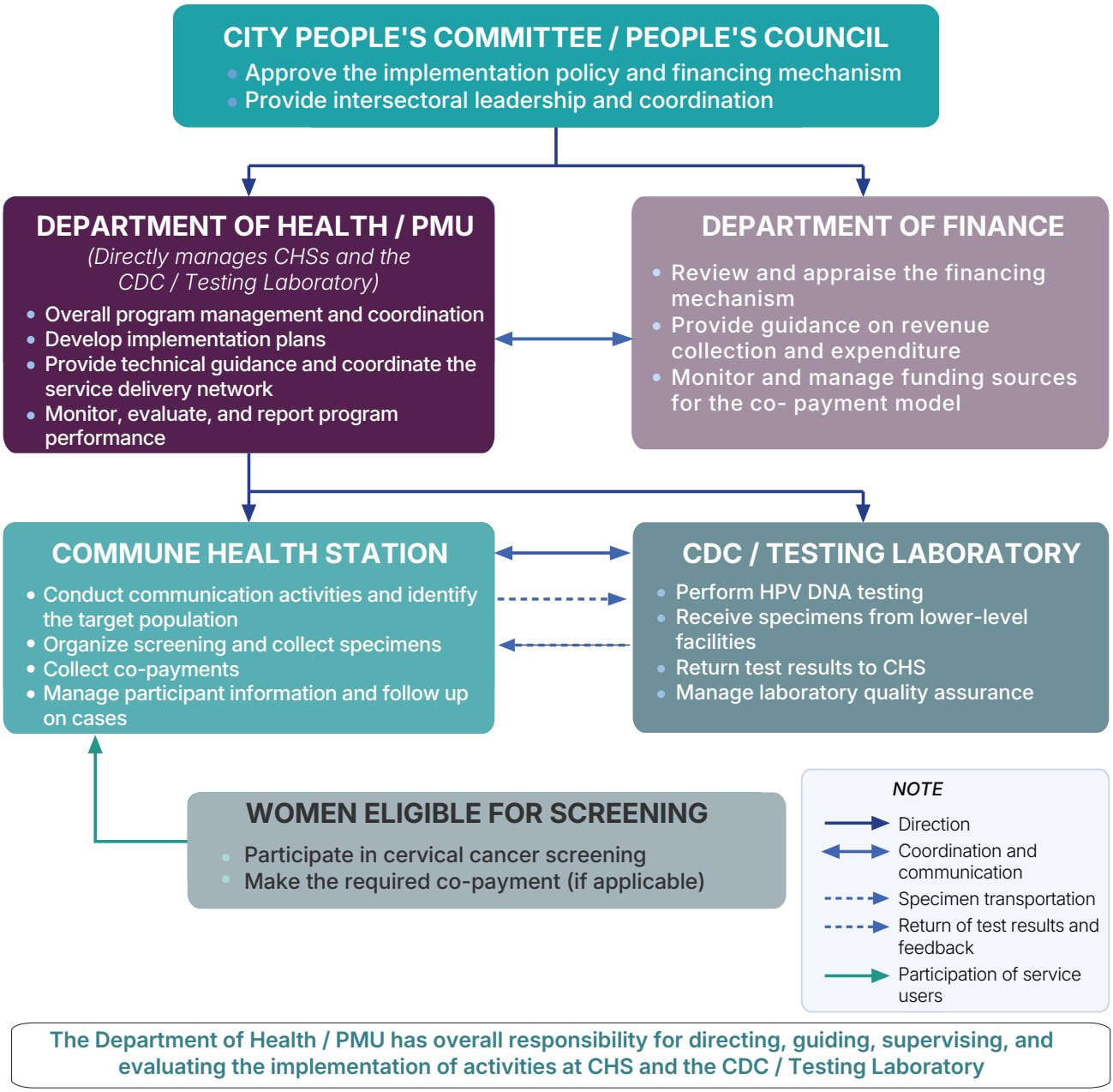


Figure 3. Implementation organizational structure

FINANCIAL FLOW MANAGEMENT

The proposed financial flow management process is illustrated in Figure 4. The roles and responsibilities of key stakeholders are summarized below:

- **CITY PEOPLE'S COMMITTEE AND PEOPLE'S COUNCIL:** Approve programme policies and implementation plans; Allocate and secure programme funding; Mobilize and coordinate financial resources; Oversee the use of public funds.
- **DEPARTMENT OF HEALTH / PROJECT MANAGEMENT UNIT:** Coordinate and oversee programme implementation; Manage programme finances, including financial reconciliation and payments; Monitor programme quality and evaluate implementation performance; Provide training and technical guidance; Develop communication strategies, implementation plans, and operational guidance.
- **CDC / TESTING LABORATORY:** Receive specimens and perform HPV DNA testing; Ensure laboratory quality assurance and quality control; Manage laboratory results and data; Provide technical support to CHSs.

- **COMMUNE HEALTH STATIONS:** Conduct community outreach and communication in collaboration with relevant stakeholders; Register participants, collect user co-payments, and maintain financial records; Collect, store, and transport specimens; Deliver test results, provide counselling, and coordinate follow-up care for women with positive results.
- **EMPLOYERS:** Encourage and support financial contributions towards HPV DNA testing for female employees.
- **WOMEN'S UNION AND LABOUR UNIONS:** Support community engagement, communication, and programme implementation activities.
- **HPV TEST SUPPLIERS:** Participate in price negotiations and provide the integrated service package, including installation and maintenance of laboratory equipment, supply of HPV testing reagents and consumables, and specimen collection kits.

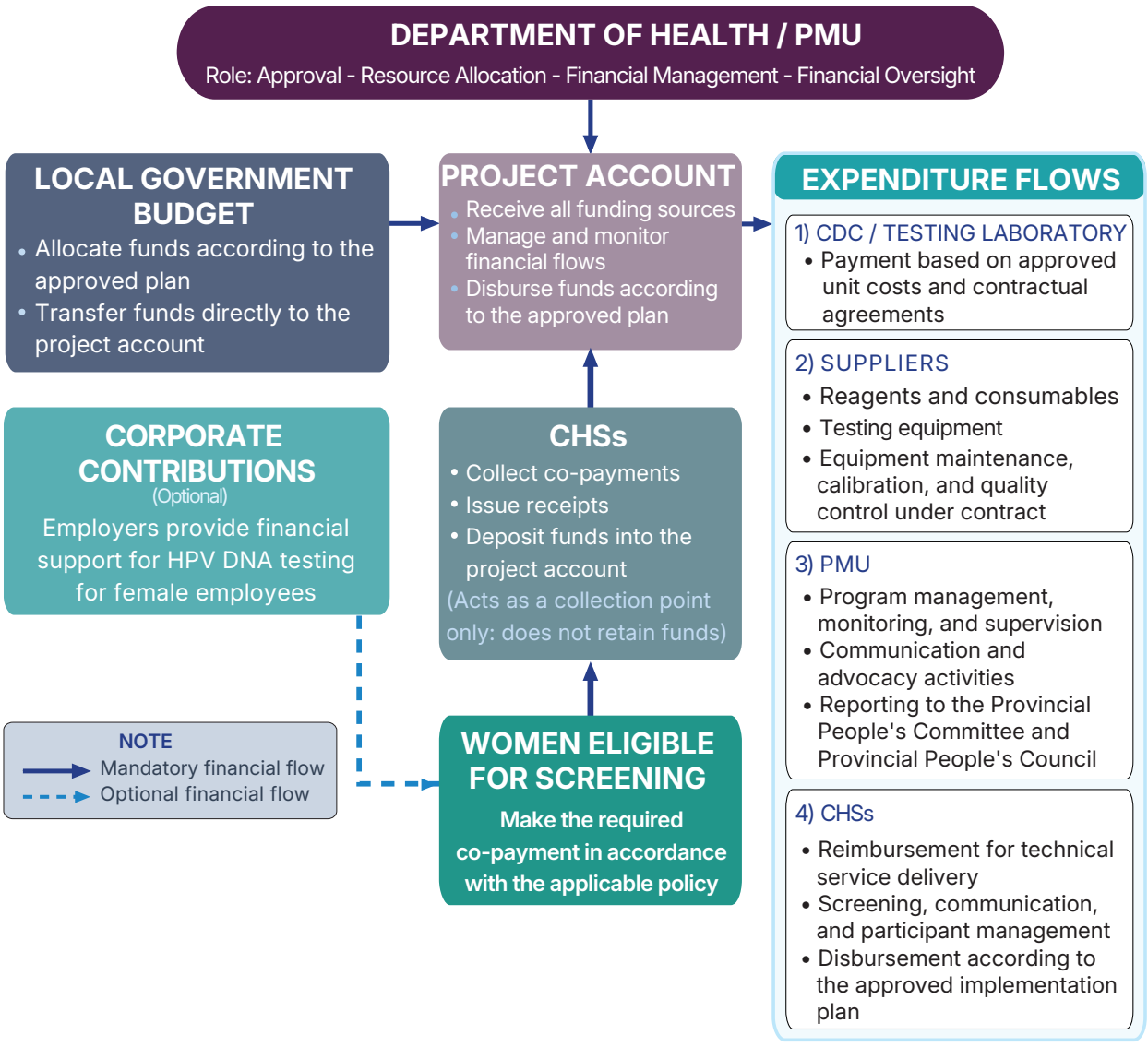


Figure 4. Overall financial flow diagram

The proposed model adopts a centralized financial management mechanism, whereby all funding from the local government and user co-payments is pooled into a dedicated programme account managed by the Department of Health / PMU. CHSs collect user co-payments and transfer the funds to the programme account in accordance with the programme's standardized financial procedures.

Programme funds are disbursed based on verified service delivery volumes and reconciled data between CHSs, the provincial CDC / testing laboratory, and the Department of Health / PMU. Funding

is allocated to all key programme components, including laboratory testing, sample collection and transportation, training, communication, programme management, monitoring, and coordination.

A performance-based payment mechanism, combined with routine reconciliation of financial and service delivery data through the programme information system, is proposed to enhance transparency, strengthen financial accountability, improve fund flow management, and promote the efficient use of programme resources.

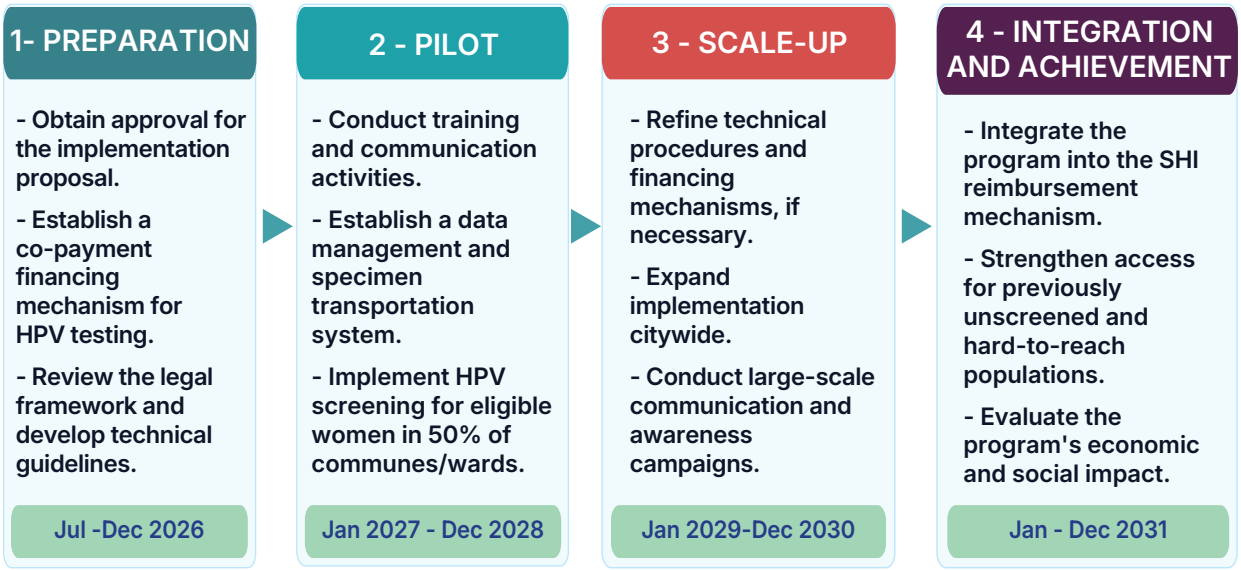


Figure 5. Roadmap for the implementation of HPV DNA testing in CxCa screening

To ensure feasibility and long-term sustainability, the programme is proposed to be implemented through a phased implementation roadmap, enabling the health system to progressively build implementation capacity while allowing adjustments to reflect the local government's fiscal capacity (Figure 5). The phased approach allows for the gradual expansion of programme coverage, continuous monitoring of implementation progress, and incorporation of lessons learned to inform timely adjustments and minimize the risks associated with large-scale implementation. It also facilitates the effective mobilization and coordination of financial and operational resources.

CONCLUSIONS

- The proposed HPV DNA testing co-payment model in Hai Phong represents a feasible interim financing solution while HPV DNA testing for CxCa screening is not yet covered by the SHI scheme. The model leverages the existing primary healthcare network by combining decentralized specimen collection at CHSs, centralized laboratory testing at the provincial CDC, and a co-payment mechanism shared between the local government and service users. This approach is expected to expand access to screening while reducing the fiscal burden on the local government.
- The analysis indicates that the proposed programme has the potential to generate substantial health and economic returns. Population-based HPV DNA screening could significantly reduce CxCa incidence, mortality, and the overall disease burden while delivering considerable socioeconomic benefits. Where adequate financial resources are available, screening all women aged 30–65 years is expected to achieve the greatest public health impact. In more resource-constrained settings, prioritizing women aged 30–49 years provides a more cost-effective implementation option. The choice of screening scale should therefore be guided by local fiscal capacity, implementation readiness, and policy priorities.
- The experience from Hai Phong could provide valuable real-world evidence to support the Ministry of Health and Viet Nam Social Security in refining sustainable financing mechanisms, and gradually including HPV DNA testing within the SHI benefit package and supporting the scale-up of the model across Viet Nam.

RECOMMENDATIONS

- ***For the Ministry of Health:*** Finalize national guidance on CxCa screening using HPV DNA testing; Develop a sustainable financing mechanism and a roadmap for the gradual inclusion of HPV DNA testing into the SHI benefit package; Develop national policies and implementation guidance for scaling up the HPV DNA testing co-payment model, drawing on evidence and lessons learned from the Hai Phong pilot to refine financing arrangements and implementation approaches.
- ***For Viet Nam Social Security:*** Collaborate with the Ministry of Health to evaluate the outcomes of the Hai Phong pilot; Assess reimbursement options and progressively integrate HPV DNA testing for CxCa screening into the SHI benefit package when the necessary policy, technical, and financial conditions are met.
- ***For the Hai Phong People's Committee:*** Approve the proposed co-payment model and allocate adequate funding for implementation during 2026–2031, selecting either Screening Scale A or Screening Scale B according to available resources and programme priorities at each stage; Direct the Integration of HPV DNA testing for CxCa screening into existing women's health programmes and promote multisectoral engagement across relevant government departments, mass organizations, and other stakeholders.
- ***For the Hai Phong Department of Health:*** Implement the proposed model based on decentralized specimen collection at CHSs and centralized HPV DNA testing at the provincial CDC; Strengthen communication, capacity building, programme monitoring, and evaluation to generate evidence for continuous improvement and future scale-up of the model.

ACKNOWLEDGEMENTS

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